

Build. Code. Deliver.

Teacher guide | Setup, teaching steps, answers and scoring

Grades 6-8 | 8 lessons | 16 classes of 45 minutes

LEGO Education Computer Science & AI Kit 6–8 (#45522, 2026)



This picture shows an idea. Use LEGO's instructions to build the real robot.

Build a rover: a small robot that moves on wheels. Help it carry a paper parcel and stop at a red line. Try it, write down what happens, and make it better.

Get ready to teach

Who is this course for?

Students in grades 6-8 work in teams of four. They should know how to open a webpage and save a file. They do not need earlier coding experience. Read directions aloud and help with measuring or averages when needed.

Test the real kit first

This is a teaching draft. These coding steps have not yet been tested on a real robot or in a classroom. Before teaching, build each program in Coding Canvas and check it on set #45522. The pictures show ideas, not exact building instructions.

Materials and setup

Use one LEGO set #45522 and one computer or tablet for each team of four. Give each student a job: build, code, start and stop the robot, or write test notes. Swap jobs after each test.

Gather a ruler, tape, flat red and white paper, light cardboard, scissors and tape that is easy to remove. Make a wide lane on a flat floor. Leave lots of empty space beyond the red finish line. Keep the lane away from stairs and table edges.

Before class, build a wheeled base using LEGO's instructions for set #45522. Fit the color sensor so it looks down at the floor, in front of the wheels. A sensor is a part that detects something. Check the real build and take a photo for students to follow. The pictures in this course show ideas; they are not step-by-step LEGO build guides.

Open code.legoeducation.com. This is Coding Canvas, LEGO's coding app. Follow LEGO's setup steps to connect the right team's robot parts. Use the lesson's coding steps to make a program: a list of instructions for the robot. Save it and test it before class. The downloaded steps are a reading guide, not a file the app can run on its own.

Charge the robot parts as LEGO directs. Lift the wheels off the floor for the first check. Test which way they turn, how to stop the program, and how to turn off the robot. Keep the computer or tablet connected while the robot runs.

Before students arrive

Check each team's robot

Charge the parts. Connect the correct robot to the correct device. Open the saved code. Raise the wheels and check which way they turn. Test the screen Stop control and how to turn the power off. With the wheels still, check red, white and another color. Then try one slow drive that stops at red.

Arrange the room

Mark a separate lane for each team. Leave extra space beyond each finish line. Label the kits. If possible, keep one working robot ready to help find problems with another team's setup. Keep each robot's test notes with its kit.

Keep students safe

Ask the teacher before each run. Start slowly. Keep one person ready at the Stop control. If the robot moves the wrong way, stop at once. If the screen control does not work, use the power-off method the teacher showed you. Test both ways to stop before students drive.

Keep fingers, hair and loose clothes away from the wheels. Turn the robot off before changing its parts. Stop if the sensor cannot name the color or the connection is lost. A red floor line tells this robot where to stop. It cannot use that color sensor to spot people, walls or table edges.

Two classes per lesson

First class: share the goal and make a prediction (5 minutes); show the first steps (10); build and code (20); try one test (5); save and tidy (5). A prediction is what you think will happen and why.

Second class: review (5 minutes); test and improve (25); explain your results to another team (10); answer one short question, save and tidy (5).

Help everyone take part

Explain words as you go

Read the lesson's "Words to know" before building. Point to the real part or show the action. For example: "This is the sensor. It detects a color." Ask a student to explain the idea in their own words. The word list at the back is there for quick help.

Make directions easy to follow

Show one step at a time. Use paper arrows for routes. Offer the audio and its written words. Let students draw and explain aloud when writing is a barrier. Give students extra time to read and measure.

Share the jobs

Use four jobs: build, code, start and stop, and write test notes. Swap after each test so everyone gets to code and explain. Check what each student understands, even when the team shares one robot.

If a team is ready for more

Add one new challenge, such as a different parcel size. Ask the team to explain what they will keep the same in their tests. More speed is not the goal.

If you do not have kits yet

Use the screen robot to practice steps in order and rules for choosing an action. Make paper parcel holders. Write "screen test" or "paper model" on the results. Later, test the real robot's movement and stopping. Success on the screen does not prove the real build works.

What to collect

At the end of each lesson, collect team test notes and one short answer from each student. At the end of the course, collect the saved code and directions another team can follow. Give credit for honest notes and useful changes, even when a run does not pass.

The eight lessons

Lesson	What students make or do	Main idea
1. Wake up your rover	Make the screen respond to a color	A color goes in; a message comes out
2. Make a short drive	Drive a little, then stop	Short drives and measuring
3. Build a parcel holder	Keep a paper parcel on the robot	Build and compare two holders
4. Help your robot read colors	Find a finish line the sensor can read	Check color readings
5. Plan the delivery route	Drive, turn and reach the delivery space	Put steps in order
6. Stop at the red line	Use a color reading to choose when to stop	Use color to choose move or stop
7. Make the delivery	Carry the parcel and stop in the right place	Test three rules for success
8. Choose a build and show it	Give your robot a new job	Explain changes and results

Plan the time

Use two classes a week for eight weeks, or four classes a week for four weeks. Each class lasts 45 minutes. Keep time for repeat tests and explanations. Add a setup class if the robots are not built yet.

Wake up your rover

What students should learn

Show a color to the sensor and explain the message that appears.

Words to know

Sensor: A part that detects something, such as a color.

Program: The steps the robot follows.

Input / output: The color reading goes in. A screen message comes out.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

Each of the three colors gives the right message. The wheels do not move.

Question and example answer

What is the input in this test? What is the output?

The input is the color reading. The output is the screen message.

Common mix-up

The robot does not know what DELIVERY means. It follows the rule you wrote.

If it does not work

Check that you connected the right robot. Make sure the sensor can see the paper. If a wheel moves, stop and ask the teacher to check the code.

Try more

Make messages that use words or symbols, so students do not have to tell screen colors apart.

Make a short drive

What students should learn

Make the robot stop after a short time. Measure how far it travels.

Words to know

Motor: A part that makes something move.

Bounded: Set to stop after a certain time or number of steps.

Trial: One test run.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

The robot stops after each of three short drives. You also show that the Stop control works.

Question and example answer

Why might the same code move the robot a different distance tomorrow?

The floor, battery charge, wheels or load may be different. Measure the distance each time instead of guessing.

Common mix-up

A speed setting of 20 does not mean 20 centimetres per second.

If it does not work

Stop. Check the charge, connection and wheel space. Test the wheel direction with the wheels raised before trying again.

Try more

Try a second low speed. Do three runs at each speed. Compare the distances.

Build a parcel holder

What students should learn

Build two versions of a holder and use your tests to choose one.

Words to know

Cargo: The thing your robot carries.

Cradle: A holder that keeps the parcel in place.

Fair test: Change one thing and keep the rest the same.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

Your holder keeps the parcel on in at least two of three runs. It does not rub a wheel or cover the sensor.

Question and example answer

Why should you keep the same parcel when you compare two holders?

A different parcel might change the result. Keeping it the same helps you test the holder itself.

Common mix-up

A taller holder is not always better. It can make the robot tip or block a part.

If it does not work

Turn the power off. Check for rubbing, loose tape or a top-heavy load. Change one part, then test again.

Try more

Give your drawing to another team. Can they make the same holder without extra help?

Help your robot read colors

What students should learn

Test paper colors and choose a setup that gives clear readings.

Words to know

Reading: The information a sensor gives you.

Unknown reading: The sensor cannot name the color.

Fair test: Change one thing at a time.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

You record nine readings, try one change and explain which paper setup you will use.

Question and example answer

The sensor names red correctly only twice in three tries. What should you do before driving?

Keep the wheels still. Check the light, paper and sensor height. Change one thing and test again.

Common mix-up

A color sensor reads colors. It does not tell you how far away a wall is.

If it does not work

Hold the paper flat. Keep the sensor steady. Reduce glare, or try another sheet of paper. Change one thing at a time.

Try more

Count how many readings were right. Would you trust the setup after only three tries? Explain why.

Plan the delivery route

What students should learn

Put steps in order, then test and adjust a short route.

Words to know

Sequence: Steps in order.

Calibrate: Test and adjust the settings.

Variable: One thing you can change, such as drive time.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

The robot stops inside your marked space in at least two of three runs.

Question and example answer

Why does one good run not prove the route will work every time?

Small changes in the floor, charge or start position can change the result. More runs help you spot a pattern.

Common mix-up

A perfect turn on a screen does not set the correct turn time for your real robot.

If it does not work

Test the straight parts and turn on their own. Check the starting direction before changing the code.

Try more

Compare two routes to the same space. Which takes fewer steps? Which works more often?

Stop at the red line

What students should learn

Make the robot move on white and stop on red or an unknown reading.

Words to know

If-then rule: If this happens, do that.

Loop: Steps that repeat.

Bounded: Set to stop after a certain time or number of steps.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

Three runs stop when the sensor reads red. Unknown readings also stop the robot. It never restarts on its own.

Question and example answer

Why might the robot miss a narrow red strip even though it reads red while still?

It moves between readings. It may pass the strip before the next reading. A wider strip or lower speed gives it more time to see red.

Common mix-up

This rule looks for a floor color. It cannot spot a person or wall in the way.

If it does not work

Stop driving. Check the colors while still, then check the strip width and code order. Ask the teacher before another floor run.

Try more

Show a message explaining the stop: RED LINE, CHECK COLOR or TIME UP.

Make the delivery

What students should learn

Join your tested parts and check the whole delivery three times.

Words to know

Success criteria: The things that must happen for a test to count as a success.

Version: One saved design or program before you change it.

Reliable: Works as expected again and again.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

At least two of three runs meet all three rules. Keep a record of every run, including the ones that fail.

Question and example answer

Your first version passed once in three runs. Your new version passed twice. What can you say?

The new version did better in these tests. You need more tests before saying it works every time.

Common mix-up

Reaching the finish does not count as a full delivery if the parcel falls off.

If it does not work

Check the holder, movement and color reading one at a time. Then try them together again.

Try more

Add a new delivery rule. Explain what you would need to change and how you would test it.

Choose a build and show it

What students should learn

Choose one of five builds, test it and explain how you improved it.

Words to know

Attachment: A part you add to the robot for a new job.

Evidence: Measurements and notes that show what happened.

Handover: Directions that help someone else use your robot.

First class: show, build and try

Use the matching workbook page. Read the new words and point to the parts. Ask “What do you think will happen, and why?” Show the first step. Let teams build and try one test. Before driving, check the lane and make sure someone is ready to stop the robot.

Second class: test and improve

Have teams repeat the test and write down every result. Change one thing and try again. Ask “What stayed the same?” and “What did your notes show?” Finish with the question below. Save the code.

What to look for

The chosen build meets its own rules in at least two of three runs. Another team can follow your directions. Explain one improvement and one remaining problem.

Question and example answer

What should you do if your robot fails during the class demonstration?

Stop it. Explain what happened. Show your earlier results and name one thing you would test next. Do not hide the failed run.

Common mix-up

A good project does not need a perfect last run. Clear, honest test notes matter too.

If it does not work

Use your saved code and setup drawing to check what changed. If needed, show one part that works and explain what still needs fixing.

Try more

Plan a different build choice for a later lesson. Explain which parts and tests would need to change.

Score the learning

A rubric is a scoring guide. Give 1 to 4 points in each row, for up to 16 points. Ask each student to explain their part. If a student needs a safety reminder, stop and teach that step again before another drive.

What to check	4: Can explain and do it	3: Mostly there	2: Needs some help	1: Needs a new start
Goal and setup	Clear rules for success. Another team can follow the setup.	Clear rules. Setup needs one small detail.	Some rules or setup steps are missing.	Cannot yet explain the goal or setup.
Explain the code	Explains the steps, choices and what the code cannot do.	Explains the steps and how the robot stops.	Explains some steps with help.	Cannot yet link the code to the action.
Test and improve	Repeats fair tests. Uses results to explain a change and what needs more work.	Repeats tests and uses the notes to choose a change.	Some tests or notes are missing, or too many things change at once.	Needs help to test and keep useful notes.
Share the work	Another team can follow the directions. Each student explains their part.	Directions are clear. Students share jobs.	Directions need detail, or jobs are not shared well yet.	Directions are hard to follow. Individual work is not clear yet.

Example feedback

“You explained your stop rule clearly. You changed both speed and marker width in your tests. We cannot tell which change helped. Keep the speed the same. Test two marker widths three times each.”

Optional extra: sort colors on the screen

Keep the wheels still. Try two colored parcel labels and a new color. Make a table that links each color reading to a screen message. Explain what happens when the color is unknown. A robot that physically sorts parcels would need a separate build and new tests.

Use the screen robot

A simulator lets students try an idea on a screen. It is not a test of the real robot. Here, a forward step moves one square and a turn is a quarter-turn.

Show these steps

Choose a map. Write where you think the robot will finish. Add instructions. Use “One step” to watch each action. Save a test result and download the notes before closing or refreshing the page.

Explain “If blocked”

This instruction checks the next square. If a wall or the edge of the grid is ahead, it turns right. Otherwise, it moves one square forward. It makes one choice each time this instruction runs. It does not drive around all walls on its own.

Example routes

F means forward one square. R means turn right. I means use “If blocked.”

First delivery: F F F R F F F F.

Around the corner: F R F F F F R F.

Sensor decision: F I F F F F.

Keep screen and floor tests separate

The screen robot checks for walls. The real kit in these lessons reads floor colors. Screen turns and distances are exact. Real wheels can slip or turn too far, so students must test and adjust them on the floor.

Show the course in an interview

What this sample shows

You can show how you planned lessons, linked activities to learning goals, wrote feedback and made teacher support materials. Show the interactive slides, images, audio and written audio words. Explain that the real kit and classroom still need testing.

A four-minute walkthrough

1. Explain who the students are and what they should be able to do.
2. Show a lesson's building steps and matching test notes.
3. Try a screen route that fails. Show how the feedback helps a student try again.
4. Show the scoring guide, word help and audio. Say what you would test with a real kit before teaching.

Describe the tools accurately

This version is a custom website. The original course was a Rise export. Showing this website alone does not show that you built these interactions in Storyline or Rise.

Sources and credits

Selected kit and current US pricing

<https://education.lego.com/en-us/products/lego-education-computer-science-and-ai/45522/>

Official #45522 build instructions

<https://www.lego.com/service/buildinginstructions/45522>

Coding Canvas

<https://code.legoeducation.com/>

Teacher Portal and LEGO lessons

<https://teach.legoeducation.com/>

LEGO Education Python reference (optional advanced route)

<https://github.com/LEGO/LEGOEducation>

SPIKE retirement and support

<https://education.lego.com/en-us/spike-update-2026/>

Original Rise course

<https://luvuniz.github.io/robotics/>

Official starting model C103: pages 24-31

<https://www.lego.com/cdn/product-assets/product.bi.core.pdf/6640383.pdf#page=24>

Kit details and US prices checked September 10, 2026. Make and test the programs in Coding Canvas before class. Audio uses Amy Sullivan's own voice clone. AI-made pictures show ideas, not checked LEGO builds. This independent course is not endorsed by the LEGO Group.

Use the build worksheets

Every module has a printable worksheet linked from its Build slide. Modules 1-7 each have two pages. Module 8 has a two-page choice plan and two pages for each of five builds. Print only the chosen build pages for each team.

Prepare the shared base

Official starting model: C103, pages 24-31 in LEGO instruction book 6640383 for set #45522. A teacher must check any changes for floor sensing and attachments on the real kit.

[Official LEGO C103 instructions, pages 24-31](#)

The LEGO model is a starting reference. Check its stability and prepare a downward-facing sensor mount and fixed attachment points. Take clear photos of the actual adapted class base. The original attachment plans in these worksheets have not been physically tested.

Before giving a worksheet to students

- [] Build and run the base yourself. Check the screen Stop and power-off methods.
- [] Measure the clear mounting area. Choose appropriate reusable connectors.
- [] Try each final option you will offer. Remove an option from the choice list if your adapted base cannot carry it safely.
- [] Prepare the paper, card, washable markers and floor protection in advance.
- [] Keep the full official manual available. The worksheet diagrams are attachment plans, not exact LEGO part diagrams.

Timing

Use the usual two 45-minute classes per module. Prepare the base before class or add an extra base-building session. For module 8, each team chooses one build. Trying all five would need extra lessons.

Teach the five final choices

Choice	Teacher check	Program to check
Parcel delivery	Low tray; light parcel; sensor clear	Lesson 6 color stop; max 20 checks
Drawing robot	Washable marker; protected floor; tip does not lift wheels	Short timed forward drive, then Stop
Paper sweeper	Soft paper only; bristles bend; wheels stay clear	Short timed forward drive, then Stop
Ball pusher	Light paper ball; wide clear lane; V not a clamp	Short timed forward drive, then Stop
Rescue carrier	Paper figure only; wide tow bar away from wheels; straight route	Short timed forward drive, then Stop

Score all five choices fairly

Use the same four-row scoring guide. Judge each build against its own three success rules. Students should record three runs, use a result to explain one change and give clear start-and-stop directions. A sweeper or ball pusher is supposed to touch its paper test object; it should not touch people or unrelated objects.

Keep the explanations accurate

Only the delivery choice uses the red-line stopping program. The other four start with a short timed drive. A marker holder draws; it does not read the drawing. A V guide pushes a ball; it does not grab it. The rescue carrier tows a paper figure; it cannot rescue a person.

Help without doing the thinking

Ask: “What should your robot do?” “Which rule did this run miss?” “What one thing could you change?” If a build struggles, stop and check weight, rubbing and attachments before changing the code.

Word help

A glossary is a list of words and their meanings. Use this page when you meet a word you do not know.

Word	Plain English meaning
Rover	A small robot that moves on wheels.
Program	A list of steps a computer or robot follows.
Coding block	A piece on the screen that gives the robot an instruction.
Sensor	A part that detects something, such as a color.
Motor	A part that uses electricity to make something move.
Input / output	Input is information going in. Output is the action or message that comes out.
Bounded	Set to stop after a certain time or number of steps.
Trial	One test run.
Fair test	Change one thing. Keep the other things the same.
Cargo / cradle	Cargo is what the robot carries. A cradle is a holder that keeps it in place.
Calibrate	Test and adjust settings so the robot does what you need.
Sequence / loop	A sequence is steps in order. A loop repeats steps.
Variable	One thing that can change, such as speed or time.
Mean	An average: add the numbers, then divide by how many numbers you added.
Success criteria	The things that must happen for a test to count as a success.
Evidence	Measurements or notes that show what happened.
Attachment	A part you add to the robot for a new job.

What will the kits cost?

These are US prices for LEGO Education Computer Science & AI 6-8 (#45522). The supply amounts are estimates: money set aside for materials.

What to buy	US dollars	What the amount covers
One kit · up to 4 learners	\$529.95	LEGO's listed price
Six individual kits · 24 learners	\$3,179.70	6 × \$529.95; fewer charging extras than bundle
Classroom bundle · 24 learners	\$3,499.00	6 kits plus charging and replacement extras
Supplies for a small tryout	\$35.00	Money set aside for: paper, tape, ruler and cargo materials
Supplies for six teams	\$120.00	Money set aside for six teams
Small tryout total before tax / shipping	\$564.95	1 kit + \$35 supplies; using a device you already have
Class total before tax / shipping	\$3,619.00	1 bundle + \$120 supplies; using 6 devices you already have

Prices were checked on LEGO's US website. Check them again before buying. The totals do not include tax, shipping, computers, school computer support or teacher training. The course and screen practice area do not need a paid subscription. No kit has been bought.

What comes with the class bundle?

The bundle has six kits, three extra multi-chargers, eight extra connection cards and a replacement brick pack. Buying six single kits costs \$319.30 less, but does not include all those extras.

Start small

Try one kit with four students and a computer or tablet you already have. For 24 students in teams of four, use six kits. For pairs, 12 kits cost \$6,359.40 before other costs. One spare kit adds \$529.95. These are kit purchases, not monthly fees.

Buying an older kit?

EV3 and Robot Inventor use different parts and coding apps. These lessons' coding steps would need changes. SPIKE direct sales ended June 30, 2026; LEGO says its app will be supported through June 30, 2031. Before buying used, check that the parts, charger and app you need are available.

[Selected kit and current US pricing](#)

[SPIKE retirement and support](#)